# Prompt 4 - neutral prompt

### System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

## Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

### User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and determine whether the explanations are similar or different based on numerical evidence.

**Assistant**

I'll help you train two GNN models, compare their performances, and explain their predictions using EdgeSHAPer. Let's start by finding a suitable dataset! 🔍

(Calling tool `find_datasets`)

**Assistant**

Great! I found a compound activity prediction dataset. Let me load it and set up the workflow. 📊

(Calling tool `load_dataset`)

**Assistant**

Perfect! I have a dataset with 8,348 compounds. Now let me split it into train/val/test partitions for GNN training. 📂

(Calling tool `split_dataset`)

**Assistant**

Dataset split complete! Now let me prepare the GNN dataset and train two different GNN models. I'll use **GCN** and **GAT** architectures for comparison. 🧪

(Calling tool `prepare_gnn_dataset`)

**Assistant**

Excellent! GNN dataset prepared with 5,342 training, 1,336 validation, and 835 test compounds. Now let me train two different GNN models: **GCN** and **GAT**. 🚀

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

**Assistant**

Both GNN models are training in the background. Let me check their progress. ⏱️

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

**Assistant**

Both models are still training. Let me check again after a moment. ⏳

(Calling tool `check_gnn_training`)
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**Assistant**

Models are still training. Let me check again. ⏳

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**Assistant**

Still training. Let me check again. ⏳

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(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

### Assistant

🎉 **GCN model completed!** Training finished with impressive results:

| Metric | Value |
|---|---|
| Test Accuracy | 94.85% |
| Best Validation Accuracy | 95.13% |
| Training Accuracy | 95.34% |
| Matthews Correlation Coefficient | 0.907 |

The GAT model is still training. Let me check on it. ⏳

(Calling tool `check_gnn_training`)

### Assistant

GAT model is still training. Let me check again. ⏳

(Calling tool `check_gnn_training`)

### Assistant

GAT model is still training. Let me check again. ⏳

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(Calling tool `check_gnn_training`)

**Assistant**

🎉 **GAT model completed!** Both models have finished training. Here's the performance comparison:

| Model | Test Accuracy | Best Val Accuracy | Training Accuracy | MCC |
|---|---|---|---|---|
| GCN | 94.85% | 95.13% | 95.34% | 0.907 |
| GAT | **95.45%** | 95.21% | 95.38% | **0.908** |

📊 **Summary**: GAT slightly outperforms GCN with a **0.6% higher test accuracy** and marginally better MCC.

Now, let me select a compound from the test set that **both models correctly predict as active**, and then explain their predictions using EdgeSHAPer. 🔬

(Calling tool `select_compound_for_edgeshaper_agreement`)

**Assistant**

Excellent! Found a compound that **both models correctly predict as active** with high confidence:

| Property | Value |
|---|---|
| SMILES | `O=C(CCCc1cn(-c2ccc([N+](=O)[O-])cc2)nn1)NCCCCN1CCN(c2cccc(Cl)c2Cl)CC1` |
| True Label | 1 (Active) ✅ |
| GCN Prediction | 1 (Active) - Confidence: 97.7% |
| GAT Prediction | 1 (Active) - Confidence: 99.8% |

Now let me load the GNN models and run **EdgeSHAPer** on this compound for both models. 🧪

(Calling tool `load_gnn_model_mcp`)
(Calling tool `load_gnn_model_mcp`)

### Assistant

Both models loaded successfully! Now let me run **EdgeSHAPer** to compute edge importance for both models on this compound. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

### Assistant

Excellent! EdgeSHAPer explanations are complete for both models. Let me visualize the results and compare them. 📊

(Calling tool `visualize_edgeshaper_results`)

### Assistant

(Calling tool `visualize_edgeshaper_results`)

### Assistant

Excellent! Now let me analyze the EdgeSHAPer results numerically and compare the explanations. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

### Assistant

(Calling tool `get_edgeshaper_bond_environments`)

### Assistant

Now I have all the data needed for a comprehensive comparison. Let me provide the complete analysis! 📊

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## 📊 Complete Analysis: GCN vs GAT Performance & EdgeSHAPer Explanations

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## 🧪 Model Performance Comparison

| Metric | GCN | GAT |
|---|---|---|
| Test Accuracy | 94.85% | **95.45%** ✅ |
| Best Validation Accuracy | 95.13% | 95.21% |
| Training Accuracy | 95.34% | 95.38% |
| Matthews Correlation Coefficient | 0.907 | **0.908** ✅ |
| Test False Positives | 102 | 79 ✅ |
| Test False Negatives | 147 | 168 |

🏆 **Winner: GAT** - Slightly better performance with fewer false positives!

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## 🔬 EdgeSHAPer Explanation Analysis

**Selected Compound:** `O=C(CCCc1cn(-c2ccc([N+](=O)[O-])cc2)nn1)NCCCCN1CCN(c2cccc(Cl)c2Cl)CC1`

- **True Label:** 1 (Active) ✅
- **Both Models Predict:** 1 (Active)
- **GCN Confidence:** 97.7%
- **GAT Confidence:** 99.8%

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## 📈 Numerical Comparison of Edge Importance

### Top 5 Bonds by |phi_sum| (Absolute Contribution)

| Rank | Bond | Structure | GCN phi_sum | GAT phi_sum | Agreement? |
|---|---|---|---|---|---|
| 1 | Bond 25 | C-C (ring) | **0.1213** | **0.1105** | ✅ Same |
| 2 | Bond 4 | C-C (single) | 0.1024 | **0.1181** | ✅ Same |
| 3 | Bond 38/36 | C-N / C-C | 0.1011 | 0.0918 | ⚠️ Different |
| 4 | Bond 27 | N-C | 0.0957 | - | ❌ Different |
| 5 | Bond 23/19 | C-N / N-C | 0.0934 | 0.0844 | ⚠️ Different |

### Key Numerical Evidence:

| Metric | GCN | GAT |
|---|---|---|
| **Top 2 bonds overlap** | Bonds 4, 25 | Bonds 4, 25 |
| **Mean | phi | (top 10)** |

| Metric | GCN | GAT |
|---|---|---|
| Max positive phi | 0.5124 (Bond 37) | 0.1057 (Bond 10) |
| Max negative phi | -0.5241 (Bond 36) | -0.0598 (Bond 66) |
| Trustworthiness | 0.9995 | 0.5675 |
| Fidelity | 0.9769 | 0.4983 |

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## 🔍 Similarities

1. **Top 2 bonds are identical** - Both models agree the most important bonds are:
   - **Bond 4** (C-C single bond connecting the alkyl chain to the pyrazole ring)
   - **Bond 25** (C-C bond in the piperazine ring)
2. **Predominantly positive contributions** - Both models show that most top bonds contribute positively to the active prediction
3. **Similar magnitude range** - The top bonds have comparable importance (0.08-0.12 range)

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## ⚠️ Differences

1. **Magnitude of extreme values:**
   - GCN has **much larger extreme values** (±0.52) vs GAT (±0.11)
   - This suggests GCN relies more heavily on specific bonds
2. **Different 3rd-5th ranked bonds:**
   - GCN: Bonds 38 (C-N), 27 (N-C), 23 (C-N)
   - GAT: Bonds 36 (C-C), 19 (N-C), 2 (C-C)
3. **Trustworthiness & Fidelity:**
   - GCN: **0.9995** trustworthiness (very reliable explanation)
   - GAT: **0.5675** trustworthiness (moderate reliability)

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## 📝 Conclusion

**The explanations are PARTIALLY SIMILAR but with notable differences:**

| Aspect | Assessment |
|---|---|
| Top 2 bonds | ✅ **Highly Similar** - Both models agree |
| Overall pattern | ✅ **Similar** - Both show positive contributions |
| | |

| Aspect | Assessment |
| --- | --- |
| Magnitude | ⚠️ **Different** - GCN has more extreme values |
| Lower-ranked bonds | ⚠️ **Different** - Different structural features |
| Explanation reliability | ⚠️ **Different** - GCN more trustworthy |

🎯 **Key Insight:** Both models agree on the **most critical structural features** (Bonds 4 and 25), suggesting these are the genuine activity-driving fragments. However, they **diverge on secondary features**, indicating different interpretations of less important structural elements. The GCN model provides more confident (higher trustworthiness) but potentially more extreme explanations, while GAT provides a more balanced but less certain explanation.



